Exp no: 07
24.9.26

Sliding Window

Aim:-

To write sliding window protocol code using python

Code:-

Sender.py

```
import time
import os
ws = int(input("Window Size"))
msg = input("Enter message: ")
frames = [(i+1, ch) for i, ch in enumerate(msg)]
i = 0
while i < len(frames):
    windows = frames[i:i+ws]
    with open("Sender_Buffer","w") as f:
        for no, data in window:
            f.write(f"{no}, {data}\n")
    print("\n Sending :", window)
    time.sleep(2)
    while not os.path.exists("Receiver Buffer"):
        time.sleep(1)
    with open("Receiver_Buffer") as f:
        ack, no = f.read().strip().split(,);
        no = int(no)
    print("Received", ack, no)
    if ack == "Ack" and no == window[-1][0]:
        i += ws
    elif ack == "NACK":
        i = no-1
```

open ("Receiver_Buffer", "w").close()
print ("\n Transmission completed")

receiver.py

```
import time,os
expected = 1
while True:
    if not os.path.exists("Sender_Buffer"):
        time.sleep(1)
        continue
    with open("Sender_Buffer") as f:
        frames = [[int(x.split(",")[0]), x.split(",")[1].
                  strip()] for x in f if x.strip()]
    if not frames:
        time.sleep(1)
        continue
    print("\n Received", frames)
    for no, data in frames:
        if no != expected:
            with open("Receiver_Buffer","w") as f:
                f.write(f"NACK,{expected}")
            break
        expected = 1-1
    else:
        with open("Receiver_Buffer","w") as f:
            f.write(f"ACK,{frame[-1][0]}")
    time.sleep()
```

O/P:-

Sender Terminal

window size: 3 Enter message: BRICKS

Sending [(1,'B'),(2,'R'),(3,'I')]

Received: ACK 3

Sending [(4,'K') (5,'S')]

Received ACK 4

Transmission completed.

Receiver Terminal :-

Received : [(1,'B'),(2,'R'),(3,'I')]

Received: [(4,'K'),(5,'S')]

Result:-

Thus the code is implemented for sliding window